ParKCa: Causal Inference with Partially Known Causes

Raquel Aoki Martin Ester
School of Computing Science - Simon Fraser Uuniversity

INTRODUCTION

This project aims to understand the leading causes of the outcome of interest in certain Computational Biology applications, which can contribute to more precise diagnosis, prognosis, and treatment of patients, based on their individual data.

- Example: drug response and adversarial drug reactions are possible outcomes of interest, and OMICS data records potential causes and confounders.

Our proposed method ParKCa uses the few known causes (prior knowledge) to learn new potential causes of certain outcome through the combination of causal discovery methods. It's main advantages are:

1. Leveraging several methods instead of using a single one can minimize biases and highlight patterns common across the methods.
2. Nice use of prior knowledge to learn new causes.
3. The aggregation of several datasets that share the same set of possible causes is easy.

Table 1. Toy example of the input data (on the top) and output data (on the bottom).

Input Data:

	Gene 1	...	Gene V	Y
Patient 1	7.39	...	1.60	0
...	...	...	...	...
Patient J	3.25	...	2.73	1

Output data:

	L_1	L_2	Known Cause
Gene 1	-1.2	-2.4	1
...	...	...	..
Gene V	0	12.3	0

Main contributions of this work:

- We introduce the problem of causal inference from observational data with partially known causes. We are the first ones to formalize it as a stacking problem.
- We propose ParKCa, a flexible method that learns new causes from the outputs of causal inference methods and partially known causes.
- ParKCa is validated on a real-world TCGA dataset for identifying genes that are potential causes of cancer metastases and on simulated genomic datasets.

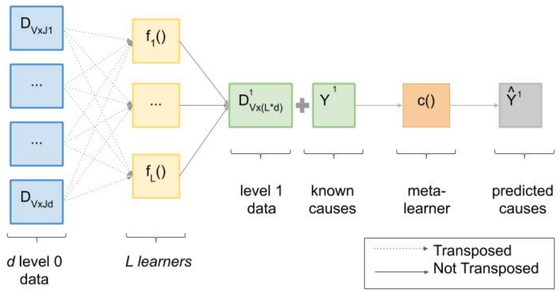

Figure 1: Illustration of our proposed method ParKCa.

METHODS

ParKCa deals with causal discovery from a stacking ensemble perspective with some adaptations:

1. Learners receive the data $D_{0(V\times J)}$ as input data along with the outcome of interest Y_0.
2. The learners' output is one value per feature $v \in \{1, ..., V\}$. The learners' outputs combined are level 1 data.
3. The meta-learner, a classifier, receives the level 1 data and as the known causes as the target.

ASSUMPTIONS:

- ParKCa assumes the number of possible causes V to be sufficiently large to fit the meta-learner.
- As an ensemble method, ParKCa also requires sufficient diversity among the learners, checked in our experiments with the Q-statistics.

LEARNERS:

To validate ParKCa, we worked with the Deconfounder Algorithm (DA)[2], BART[4], and CEVAE[5] in our experiments. These methods are causal discovery methods and/or models that estimate treatment effects.

META-LEARNERS:

We tested two PU-learning classification models, and compared their results with traditional classification models (Logistic Regression (LR), Random Forest (RF), and Neural Network (NN)). A majority vote ensemble from the methods described above and a random model is also used.

EXPERIMENTS

Genome-wide association studies (GWAS) **Datasets**

- Real-world dataset: The Cancer Genome Atlas (TCGA) dataset has available the gene expression (RNA-seq) of patients with cancer.
- Simulated datasets: We simulated GWAS data according to Want et al. [2]. We simulated ten independent datasets.

VALIDATION:

1. We evaluated all the learners adopted and ensured they were well fitted, the suppositions were satisfied, and there was enough diversity among the learners.
2. Check ParKCa contribution: we compare the number of causes identified by the learners (causal discovery or causal inference models) *versus* the number of causes identified by the meta-learner using the ParKCa methodology.

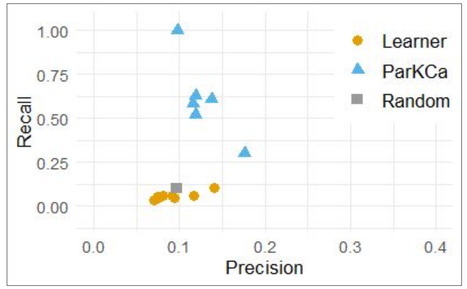

Figure 2: (*Real-world dataset*) Comparison between learners and ParKCa meta-learners to recover causes

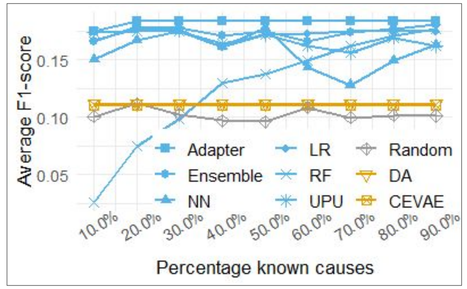

Figure 3: *(Simulated datase)* Comparison to identify causes with different proportions of known causes.

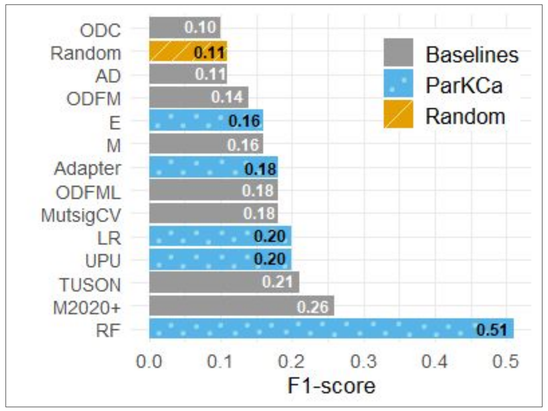

Figure 4: *(Real-world dataset)* Comparison between driver gene discovery baselines [3] and ParCKa.

CONCLUSION

ParKCa exploits partial knowledge of causes, is flexible, robust, easy to use, and demonstrates promising results on real-life and simulated datasets.

For small percentages of known causes, Adapter-PU was the meta-learner with the best performance on simulated experiments. There was almost no difference between PU models and traditional classification models when the percentage of known causes was above 70%.

REFERENCES

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